THE COOL SOLUTION

Fresno's Housing Authority keeps energy expenditures low and economic savings high with its new evaporative coolers.

by ELKA KARL

Maintenance Manager Steve Marta helps with the installation of a new Breezair evaporative cooler on a Fresno Housing Authority duplex.

Steve Marta, Maintenance Manager for the Fresno Housing Authority, wanted to change people's opinions about evaporative coolers. He felt that there was an unreasonable stigma attached to the newer energy-efficient version of these cooling units, and that people believed these units were an ineffective way to cool their homes. He also knew—given the number of complaints that his office received on hot summer days—that the conventional swamp coolers currently used on his residents' homes weren't doing the job. As a result, Marta spearheaded a project to choose new evaporative coolers as the Fresno Housing Authority's replacement product. And, thanks to the rebate program offered by Pacific Gas and Electric Company (PG&E), the Housing Authority was able to choose a top-of-the-line cooler while receiving around $100,000 worth of rebates.

The Fresno Housing Authority oversees 1,100 units of public housing, most of which are duplexes and single family units around 800-1,200 ft^2 total in size. The Housing Authority chose new efficient evaporative cooling as its replacement product for several reasons. The coolers are a considerable energy cost saving for residents when compared with traditional A/C units, which were a considered option, and the energy costs are also less when compared with older swamp coolers. The new coolers are also very effective at producing cold air, as Steve Marta and his crew observed last year during the hot August and September months in California's Central Valley, where Fresno is located.

The evaporative cooler unit selected by the Fresno Housing Authority was the Breezair. They chose the Breezair unit because the company that manufactures the product contacted the Housing Authority and offered to install a few units for free.

"We didn't just jump in and use Breezairs on HUD housing," Marta says, wanting to make this point clear. The Fresno Housing Authority had a history with the Breezair model for a year, and were able to study the evaporative cooler and determine its effectiveness. They noted that the aspen pad material in the pre-existing swamp cooler didn't hold up very well, and that it also didn't have a pump-down water removal system. The Breezair, on the other hand, has a honeycomb pad system with pads that have the potential to last up to six years. This new-style medium has a self-cleaning function that not only yields filtered air to the home, but also provides cooler air for a given set of conditions of dry-bulb temperature and humidity. Marta and his crew decided that the maintenance and labor savings would be much larger with the Breezair units when compared to the preexisting coolers. The savings could be attributed to many things, including the lower maintenance of the pads and the fact that the Breezair doesn't have water distribution tubes, which often become plugged and require service calls. Also, since the Breezairs don't need to be cleaned out or shut down at the end of the cooling season, but simply need to have all the water drained out of them, the maintenance, in general, is minimal as compared to a conventional swamp coolers. For these reasons, the Housing Authority decided to switch to the Breezair because it was the more cost-effective of the coolers. Although there were several hundred dollars of price difference between the preexisting cooler and the Breezair unit, the rebate offered by PG&E paid for most of the incremental cost, making the Breezair cooler as affordable as an older swamp cooler.

Since the Fresno Housing Authority has to deal with constant maintenance, it made sense to go with a higher cost product that required less maintenance. The new units have, in the past ten months, required no maintenance calls. With the old coolers, Marta's maintenance crews were out in the field, constantly dealing with broken, malfunctioning, or underperforming units. They averaged between 20 and 30 calls a day on the old swamp coolers.

Good Design, Great Performance

The Breezair was a definite upgrade from the old swamp coolers. It features an all-plastic design, with 4-inch-thick pads that provide even distribution of water over the pad, which increases the unit's effectiveness. The curved-rib feature on the pad frames eliminates the possibility of water spraying outside the unit. The pump, which has a 3.6-gallon capacity, is the smallest in the industry. It also ensures a steady inflow of water, which lessens deposit buildup on the pads.

The plastic exterior of the Breezair is made of stabilized, UV-resistant, structural polymer. While Marta says that the Housing Authority will have to see if the exteriors hold up over time (with constant exposure to the hot Fresno sun), the polymer is the same material used to make acid baths, battery cases, and parts of space satellites. Unlike the metal housing used in conventional evaporative coolers, the plastic won't rust or corrode, and the plastic also generates less vibration than metal covers and conducts less noise into the ductwork.

The Breezair's blower wheel is a forward-curved centrifugal fan of double width, double inlet design, with blades that are of aerofoil shape and staggered to provide smooth airflow with reduced noise levels. The Breezair also features a quiet, lightweight fan, a water spreader that provides even water distribution over the pads, and has support legs located in the support frame for easy roof installation.

A Day in the Life

In May, two representatives from PG&E, Lance Elberling, senior program engineer, and Sue Fisher, senior program manager, along with myself from *Home Energy*, headed to Fresno to present a rebate check to the Fresno Housing Authority, as well as to stop by and see how the new whole-house evaporative cooling units were working. During a conversation with Maintenance Manager Marta, he revealed that the Housing Authority may be installing at least 100 more units in its housing complexes, which seems like a good indication that things are going well.

While in Fresno, Elberling and Fisher made a pit stop at FASCO, the largest evaporative cooler distributor in the city that carries all evaporative cooler brands at its store. PG&E came down to the store to talk to the counter sales staff about evaporative coolers, as well as to distribute a display containing a PG&E fact sheet and rebate program applications to be placed next to the register for customers or contractors to take. The fact sheet, "Energy-Efficient Ducted Evaporative Coolers," is a straightforward and informative source for both contractors and homeowners who may be looking for cooling solutions (see Figure 1). Also, Elberling and Fisher educated the counter staff about energy efficient whole-house evaporative coolers and the rebate program, hoping that their increased knowledge will influence the contractors who come into the store. While talking to the staff, the FASCO store manager commented that 20 years ago, the Housing Authority would have bought the cheapest possible equipment and then replaced it every two to three years, whereas now it is buying top-quality equipment, which makes more economic sense.

Selection and Sizing

Cooler Capacity	Average Climate	Hot, Dry Climate
3,000 cfm	1,000 sq ft	750 sq ft
3,500 cfm	1,165 sq ft	875 sq ft
4,000 cfm	1,330 sq ft	1,000 sq ft
4,500 cfm	1,500 sq ft	1,125 sq ft
5,000 cfm	1,665 sq ft	1,250 sq ft
5,500 cfm	1,830 sq ft	1,375 sq ft
6,000 cfm	2,000 sq ft	1,500 sq ft
6,500 cfm	2,165 sq ft	1,625 sq ft

Note: When the floor area of your home is between sizes shown in the table, select the next larger cooler capacity.

Example 1

If you live in a hot, dry climate (such as California Energy Commission (CEC) climate zones 11, 12 or 13, the Central Valley) and the floor area of your home is 1,200 square feet, you would select a cooler capacity of 5,000 cubic feet per minute (cfm).

Example 2

If you live in an average climate, (such as climate zones 1-5 or 16, coastal, high altitude) and the floor area of your home is 1,200 square feet, you would select a cooler capacity of 4,000 cubic feet per minute (cfm).

PG&E

Figure 1. PG&E's "Energy-Efficient Ducted Evaporative Coolers" fact sheet helps customers choose the best evaporative cooler for their house and climate.

After we left FASCO we stopped by a housing authority unit at North Sherman Court in Fresno, where Marta and his crew were switching out old swamp coolers for the new Breezairs.

"The maintenance department would usually be inundated with calls, but ever since we installed the Breezairs we haven't had these calls," commented Susan Cuellar, the assistant director of Housing Services, as we stood in front of the unit on Sherman Court. Above our heads, on the hot roof, the maintenance team switched out an old cooler with the new Breezair on a one-story single-family duplex home. It takes Steve's crew about 45 minutes to switch out a unit or up to an hour if there are a few complications.

The air temperature on the day of our visit hovered in the high 90s, with the shingle temperature on the roof at 161°F. We walked into the small, neat duplex after the Breezair had been installed. Impressively, only seven minutes after the Breezair was installed, the supply air temperature delivered into the living space came down to 63°F when there was a 103°F inlet temperature.

One resident of the duplex, an older woman making tamales, commented that now they won't have to water the pads, as they had to with their old swamp cooler. She predicts that there also won't be unsightly stains and mess as there were with the other cooler, when the water ran down the roof of the house and got all over the porch.

"Will it feel more like an air conditioner now?" she asked through a translator. She was told that it would. And truly, it does feel good in the home. Compared to the stifling heat outside, the interior of the house is already almost too chilly. I looked at my arm and realized that it was covered with goosebumps.

Marta confirmed that this isn't an unusual reaction. In fact, many of the residents are actually turning the Breezairs off because they get too cold. "Anytime people are turning [the Breezairs] off or operating the unit on low speed in the summertime, you know they work," says Marta.

The day came to an end when Fisher presented a check for $95,100 at a meeting of the board of directors of the Housing Authority that evening. When the 2005 rebate funding went into effect, customers, like the Housing Authority, who had purchased and installed evaporative coolers in 2004 were able to apply for rebates. In the case of the Housing Authority, a request for 317 of the high efficiency whole-house evaporative coolers was submitted. Fisher told the board of the Housing Authority that the installation of these evaporative coolers, which operate at one-third the cost of conventional A/C units, is a boon to their tenants. "This is only going to benefit your tenants by giving them more energy efficiency and comfort," she said. In all, PG&E will refund a total of $97,500 to the Fresno Housing Authority for 325 units.


ELKA KARL

A generous rebate from PG&E helped Fresno Housing Authority to replace the inefficient, ineffective swamp coolers with new Breezairs.

However, the most critical part of this story is the human equation. The Housing Authority ran into a few problems with the project, including starting out using a few thermostats with the new evaporative coolers that had to be changed to hi/lo/off switches to simplify the unit's operation, especially for tenants who were non-English speaking. And while problems like this have cropped up occasionally, overall the residents have been able to use their new coolers effectively—and efficiently. Perhaps most importantly, the residents have also been able to use the coolers to keep themselves comfortable in a region of California where triple-digit temperatures are nearly a daily occurrence in the summertime. Overall, it seems this is a winning situation for everyone. PG&E benefits from an energy-efficient product that puts less strain on the grid, the Fresno Housing Authority profits from rebates, and the residents benefit from inexpensive, effective evaporative cooling.

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